# Validation of the GKP simulator (`simulator/sim.py`)

November 2026

This note defines the model that `sim.py` simulates and derives every reference value used to test it. For each one it says where the value comes from (a closed form or a published result), and then compares the simulator against it. Everything can be reproduced from the repo root with

```
python -m experiments.validate_sim
```

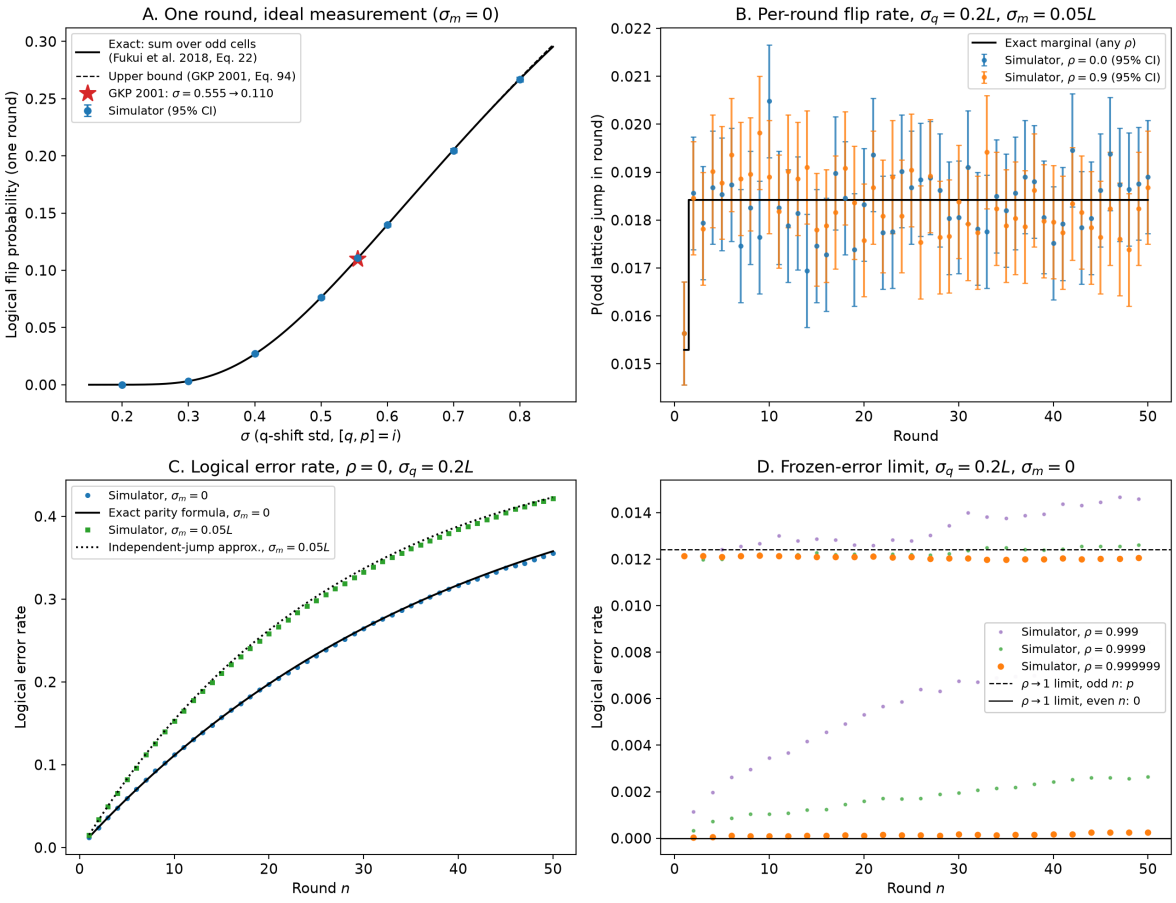



Figure 1: Simulator (markers) against closed-form references (lines). A: single round, ideal measurement. B: per-round flip rate. C: logical error rate for $\rho = 0$. D: frozen-error limit $\rho \to 1$.

# 1 Model

**Conventions.** We use dimensionless quadratures with $[\hat{q}, \hat{p}] = i$ ($\hbar = 1$) and the square GKP code, which has stabilizer $S_q = e^{2i\sqrt{\pi}\,\hat{q}}$ and lattice spacing

$$L = \sqrt{\pi}.$$

Logical $|\bar{0}\rangle$ has position peaks at $q = 2nL$, and $|\bar{1}\rangle$ has them at $q = (2n+1)L$. A shift by $2L$ is therefore a stabilizer (no effect), a shift by $L$ is the logical $\bar{X}$, and a shift with $|r| < L/2$ is correctable. This is the convention of Gottesman–Kitaev–Preskill [1]. Throughout, $\sigma$ means the standard deviation of the physical $q$-shift in this convention (see Section 5 for the factor $\sqrt{2}$ in Grimsmo–Puri [3]).

**Noise** (`simulator/noise.py`, `NoiseModel`). For each trial and round $t = 1, \ldots, T$:

$$\varepsilon_1 \sim \mathcal{N}(0, \sigma_q^2), \qquad \varepsilon_t = \rho\,\varepsilon_{t-1} + \eta_t, \quad \eta_t \sim \mathcal{N}\big(0, \sigma_q^2(1-\rho^2)\big),$$
$$\nu_t \sim \mathcal{N}(0, \sigma_m^2) \text{ i.i.d.}$$

Hence $\operatorname{Var}\varepsilon_t = \sigma_q^2$ for every $t$ and every $\rho \in (-1, 1)$; the AR(1) process is stationary.

**Error-correction round** (`Simulator.run`). Starting from residual $x_0 = 0$:

$$\begin{aligned} x_t^{\mathrm{pre}} &= x_{t-1} + \varepsilon_t && \text{(displacement error)} \\ s_t &= \operatorname{wrap}(x_t^{\mathrm{pre}} + \nu_t) && \text{(noisy syndrome)} \\ c_t &= \mathcal{D}(s_1, \ldots, s_t) && \text{(decoder)} \\ x_t &= x_t^{\mathrm{pre}} - c_t && \text{(correction)} \end{aligned}$$

with $\operatorname{wrap}(y) = y - L\,\operatorname{round}(y/L) \in [-L/2, L/2)$.

**Logical error after round $t$.** Write $x_t = k_t L + r_t$, where $k_t = \operatorname{round}(x_t/L)$ and $|r_t| < L/2$. A single ideal final correction removes $r_t$, while $k_t L$ applies $\bar{X}^{k_t}$. Hence

$$\text{logical error after round } t \iff k_t \text{ is odd.}$$

The logical error rate is $P_{\mathrm{L}}(t) = \Pr[k_t \text{ odd}]$ (`SimResult.logical_error_rate`), and the lattice jump in round $t$ is $\Delta k_t = k_t - k_{t-1}$ (`SimResult.lattice_jumps`). The *nearest-lattice decoder* sets $c_t = s_t$.

# 2 Closed-form results for the nearest-lattice decoder

Let $\Phi$ be the standard normal CDF. For $Z \sim \mathcal{N}(0, \sigma^2)$ define

$$\boxed{p(\sigma) = \Pr\big[\operatorname{round}(Z/L) \text{ odd}\big] = \sum_{j \text{ odd}} \left[\Phi\left(\frac{(j+\frac{1}{2})L}{\sigma}\right) - \Phi\left(\frac{(j-\frac{1}{2})L}{\sigma}\right)\right]} \tag{P}$$

(`sim.odd_jump_probability`). The cells $|j| \geq 3$ contribute exponentially little, so

$$p(\sigma) \approx 2\big[1 - \Phi(L/2\sigma)\big] = \operatorname{erfc}\left(\frac{L}{2\sqrt{2}\,\sigma}\right),$$

and this tail is also an upper bound on $p(\sigma)$.

**Proposition 1** (Single round, ideal measurement)**.** *If $\sigma_m = 0$, a single round ends in a logical error with probability $p(\sigma_q)$.*

*Proof.* $x_1^{\mathrm{pre}} = \varepsilon_1$ and $c_1 = \mathrm{wrap}(\varepsilon_1)$, so $x_1 = L\ \mathrm{round}(\varepsilon_1/L)$ and $k_1 = \mathrm{round}(\varepsilon_1/L)$. □

*Literature.* (P) is the incorrect-decision probability of Fukui et al. [2], Eq. (22), at cutoff $v_{\mathrm{up}} = 0$. GKP [1], Eq. (94), gives the bound $p_e < \mathrm{erfc}\big(\sqrt{\pi}/(2\sqrt{2}\sigma)\big)$ and evaluates it as 0.110 at $\sigma \approx 0.555$. Both papers use the same $\sigma$ convention and the same $\sqrt{\pi}$ lattice as this note.

**Lemma 2** (Residual after a nearest-lattice correction)**.** *For every noise realisation,*

$$x_t = L\ \mathrm{round}\big((x_t^{\mathrm{pre}} + \nu_t)/L\big) - \nu_t.$$

*Proof.* $x_t = x_t^{\mathrm{pre}} - \mathrm{wrap}(x_t^{\mathrm{pre}} + \nu_t) = x_t^{\mathrm{pre}} - (x_t^{\mathrm{pre}} + \nu_t) + L\ \mathrm{round}\big((x_t^{\mathrm{pre}} + \nu_t)/L\big)$. □

After every round, then, the oscillator sits on a lattice point shifted by minus that round's measurement error. When $|\nu_t| < L/2$ (almost surely here, since $\sigma_m = 0.05L$), we have $k_t = \mathrm{round}\big((x_t^{\mathrm{pre}} + \nu_t)/L\big)$.
*Status.* The lemma follows from the model by direct algebra. We did not find it stated in this form in [1–3]. Grimsmo–Puri [3] (Fig. 10, Sec. III.2) describe the related fact that a measurement error leaves a small residual displacement after correction.

**Proposition 3** (Per-round flip probability, any $\rho$)**.**

$$\Delta k_t = \mathrm{round}(z_t/L), \qquad z_t = \varepsilon_t + \nu_t - \nu_{t-1} \quad (\nu_0 \equiv 0),$$

*and therefore*

$$\Pr[\Delta k_t \ \textit{odd}] = \begin{cases} p\big(\sqrt{\sigma_q^2 + \sigma_m^2}\big), & t = 1, \\ p\big(\sqrt{\sigma_q^2 + 2\sigma_m^2}\big), & t \geq 2. \end{cases}$$

*Proof.* Substitute $x_{t-1} = k_{t-1}L - \nu_{t-1}$ (Lemma 2) into $x_t^{\mathrm{pre}}$; the $k_{t-1}L$ term is a lattice multiple and drops out of the rounding. The variable $z_t$ is a sum of independent Gaussians: $\varepsilon_t$ has variance $\sigma_q^2$ for every $\rho$, and $\nu_t$ and $\nu_{t-1}$ are independent of it. □

This describes the *marginal of a single round* only. When $\rho \neq 0$, the jumps $\Delta k_t$ in different rounds are correlated.

**Proposition 4** (Logical error rate for independent jumps)**.** *If the parities of $\Delta k_1, \ldots, \Delta k_n$ are independent, with odd probabilities $p_1, \ldots, p_n$, then*

$$P_{\mathrm{L}}(n) = \tfrac{1}{2}\Big[1 - \prod_{t=1}^{n}(1 - 2p_t)\Big].$$

*Proof.* $k_n = \sum_t \Delta k_t$, so $\mathbb{E}[(-1)^{k_n}] = \prod_t \mathbb{E}[(-1)^{\Delta k_t}] = \prod_t (1 - 2p_t)$. □

This is the standard parity ("piling-up") identity for independent bits. It applies to our model as follows.

- **Exact** when $\rho = 0$ and $\sigma_m = 0$: the $z_t = \varepsilon_t$ are i.i.d., so $P_{\mathrm{L}}(n) = \frac{1}{2}\big[1 - (1 - 2p(\sigma_q))^n\big]$.
- **Approximate** when $\rho = 0$ and $\sigma_m > 0$: $z_t$ and $z_{t+1}$ share $\nu_t$ with opposite signs, so $\mathrm{corr}(z_t, z_{t+1}) = -\sigma_m^2/(\sigma_q^2 + 2\sigma_m^2) \approx -0.056$ for the default parameters. We use $p_1 = p\big(\sqrt{\sigma_q^2 + \sigma_m^2}\big)$ and $p_{t\geq 2} = p\big(\sqrt{\sigma_q^2 + 2\sigma_m^2}\big)$.

- **Not valid** when $\rho \neq 0$. We claim no closed form there.

**Proposition 5** (Frozen-error limit $\rho \to 1$, $\sigma_m = 0$)**.** *If $\varepsilon_t \equiv \varepsilon$ for all $t$, every round jumps by* round$(\varepsilon/L)$*. Then $k_n = n$* round$(\varepsilon/L)$*, and*

$$P_{\mathrm{L}}(n) = \begin{cases} p(\sigma_q), & n \text{ odd}, \\ 0, & n \text{ even}. \end{cases}$$

The simulator requires $|\rho| < 1$, so we check this as a limit, letting $\rho$ approach 1.

## 3 Results

Unless noted otherwise, the parameters are $\sigma_q = 0.2L$ and $\sigma_m = 0.05L$, with 200 000 trials for check A, 50 000 trials for checks B–D, and seed 2026. The score is $z = (\hat{p} - p)/\sqrt{p(1-p)/N}$, computed with the reference value $p$.

| Check | Case | Simulated | Reference | $z$ | Type |
|---|---|---|---|---|---|
| A: single round (Prop. 1) | $\sigma = 0.200$ | 0.00001 | 0.00001 | −0.64 | exact |
| | $\sigma = 0.300$ | 0.00298 | 0.00314 | −1.21 | exact |
| | $\sigma = 0.400$ | 0.02739 | 0.02672 | +1.87 | exact |
| | $\sigma = 0.500$ | 0.07616 | 0.07632 | −0.26 | exact |
| | **$\sigma = 0.555$** | **0.11076** | **0.11031** (GKP: 0.110) | +0.64 | exact + lit. |
| | $\sigma = 0.600$ | 0.13967 | 0.13965 | +0.01 | exact |
| | $\sigma = 0.700$ | 0.20458 | 0.20535 | −0.86 | exact |
| | $\sigma = 0.800$ | 0.26687 | 0.26707 | −0.20 | exact |
| B: per-round flip (Prop. 3) | $\rho = 0$, round 1 | 0.01564 | 0.01529 | +0.63 | exact |
| | $\rho = 0$, mean of rounds 2–50 | 0.01837 | 0.01842 | −0.66 | exact |
| | $\rho = 0.9$, round 1 | 0.01564 | 0.01529 | +0.63 | exact |
| | $\rho = 0.9$, mean of rounds 2–50 | 0.01834 | 0.01842 | −1.01 | exact |
| C1: $P_{\mathrm{L}}$, $\rho = 0$, $\sigma_m = 0$ (Prop. 4) | worst of 50 rounds (47) | 0.34344 | 0.34669 | −1.53 | exact |
| | round 50 | 0.35562 | 0.35783 | −1.03 | exact |
| C2: $P_{\mathrm{L}}$, $\rho = 0$, $\sigma_m = 0.05L$ | round 50 | 0.42174 | 0.42298 | −0.56 | approx. |
| D: $\rho \to 1$, $\sigma_m = 0$ (Prop. 5) | $\rho = 0.999$, odd / even $n$ | 0.01330 / 0.00561 | 0.01242 / 0 | | limit |
| | $\rho = 0.9999$, odd / even $n$ | 0.01230 / 0.00174 | 0.01242 / 0 | | limit |
| | $\rho = 0.99999$, odd / even $n$ | 0.01207 / 0.00052 | 0.01242 / 0 | | limit |
| | $\rho = 0.999999$, odd / even $n$ | 0.01207 / 0.00015 | 0.01242 / 0 | | limit |

**A (panel A).** All eight points lie within 2 standard errors of the reference. At $\sigma = 0.555$ the exact sum (P) and the GKP bound agree to four decimals (0.1103). The bound also counts the even cells $|j| \geq 2$, but they add only $1.7 \times 10^{-6}$. This reproduces GKP's quoted value of 0.110.

**B (panel B).** Round 1 gives the same number for both values of $\rho$. This is by design: the two runs share a seed, and $\varepsilon_1$ does not depend on $\rho$. The pooled $z$ for rounds 2–50 assumes the rounds are independent, which they are not, so treat it as indicative only. Taken round by round, the most extreme of the 49 rounds has $|z| = 3.42$ for $\rho = 0$ (Bonferroni $p = 0.03$) and $|z| = 2.32$ for $\rho = 0.9$ ($p = 0.99$). We reran $\rho = 0$ with three more seeds and got a maximum $|z|$ of 2.60, 3.30 and 2.96, while the mean over rounds 2–50 stayed between 0.01835 and 0.01840 (exact value 0.01842). We read the large single-round values as the result of picking the most extreme of 49 rounds, not as a bias.

**C (panel C).** With $\sigma_m = 0$ the simulator follows the exact curve for all 50 rounds. With $\sigma_m > 0$ it lies slightly *below* the independent-jump approximation, which is the predicted direction: $z_t$ and $z_{t+1}$ are negatively correlated, so jumps tend to come in $+1, -1$ pairs that cancel in parity.

**D (panel D).** The odd-round rate approaches $p(\sigma_q) = 0.0124$ and the even-round rate approaches 0. The leftover even-round rate falls by about $\sqrt{10}$ for every factor of 10 reduction in $1 - \rho$. That fits a picture in which $\varepsilon_t$ drifts by about $\sigma_q\sqrt{1-\rho^2}$ per round.

**Conclusion.** The simulator reproduces every closed-form and literature value in this note within statistical error. Where only an approximation or a limit is available, it deviates in the predicted direction and converges as predicted.

## 4 What is *not* validated here

- For $0 < \rho < 1$ there is **no closed form** here for the logical error rate over many rounds. Those curves, for example the drop from 0.425 at $\rho = 0$ to 0.247 at $\rho = 0.9$ after 50 rounds, come from the simulator alone. They are consistent with Propositions 3–5, which cover the per-round marginal and the two ends $\rho = 0$ and $\rho \to 1$.
- Decoders other than nearest-lattice and no correction.
- Anything beyond the classical shift model: finite-energy ($\Delta$) effects, $p$-quadrature noise, ancilla noise in the extraction circuit, and loss. `quantum/encoding.py` is the starting point for checking the shift model against QuTiP.

## 5 Sources and conventions

| Ref. | Used for | Location | $\sigma$ convention |
|---|---|---|---|
| [1] GKP 2001 | Bound; value 0.110 at $\sigma = 0.555$ | Sec. VIII, Eqs. (94)–(95) | std of physical $q$-shift, $[q, p] = i$, $\hbar = 1$ (same as here) |
| [2] Fukui et al. 2018 | Exact odd-cell sum (P) | Sec. III.2, Eq. (22), $v_{\text{up}} = 0$ | std of $q$-deviation, $\sqrt{\pi}$ lattice (inferred; not stated) |
| [3] Grimsmo & Puri 2021 | Code conventions; residual-shift picture | Eqs. (4), (5a), (26); Sec. III.2; Fig. 10 | $\sigma$ per component of $\zeta$ in $D(\zeta)$; **physical $q$-std is $\sqrt{2}\,\sigma$** |

We derived Lemma 2 and Propositions 3–5, with their proofs, from the model in Section 1. We did not find them stated in this form in [1–3]. The equation and section numbers were read from the arXiv HTML versions listed below (GKP v3; Fukui v3, of which only the first half was read verbatim; Grimsmo–Puri v1). Check them against the journal versions before citing.


[1] D. Gottesman, A. Kitaev, J. Preskill, "Encoding a qubit in an oscillator," *Phys. Rev. A* **64**, 012310 (2001). https://arxiv.org/abs/quant-ph/0008040

[2] K. Fukui, A. Tomita, A. Okamoto, K. Fujii, "High-threshold fault-tolerant quantum computation with analog quantum error correction," *Phys. Rev. X* **8**, 021054 (2018). https://arxiv.org/abs/1712.00294

[3] A. L. Grimsmo, S. Puri, "Quantum error correction with the Gottesman–Kitaev–Preskill code," *PRX Quantum* **2**, 020101 (2021). https://arxiv.org/abs/2106.12989